

An algorithm for Aubert–Zelevinsky duality

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(joint work with Alberto Mínguez)

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Workshop Representation theory of p -adic groups

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An involution for finite reductive groups

Unipotent characters of $\mathrm{GL}_n(\mathbb{F}_q)$ are parametrized by partitions of n .

There is a natural involution given by transpose of partitions.

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$\mathrm{GL}_n(\mathbb{Q}_p)$		$\mathrm{GL}_n(\mathbb{F}_q)$
Aubert–Zelevinsky	$\longleftrightarrow$	Alvis–Curtis
Multisegments	$\longleftrightarrow$	Partitions
Mœglin–Waldspurger	$\longleftrightarrow$	Transpose

Aubert–Zelevinsky involution

Setting:

- F a p -adic field with norm $|\cdot|$
- G a p -adic group

Representations:

- $\text{Rep}(G)$ = smooth complex finite-length representations
- $\text{Irr}(G)$ = irreducible representations
- $\text{Cusp}(G)$ = irreducible cuspidal representations

Functors: For a parabolic subgroup $P = MN$,

$$\mathbf{i}_P^G: \text{Rep}(M) \rightarrow \text{Rep}(G), \quad \mathbf{r}_P^G: \text{Rep}(G) \rightarrow \text{Rep}(M)$$

(normalized parabolic induction and restriction).

Operator on the Grothendieck group

Let $R(G)$ be the Grothendieck group of $\text{Rep}(G)$.

Definition

Define the operator

$$D_G: R(G) \longrightarrow R(G)$$

by

$$D_G(\pi) := \sum_{\substack{P=MU \\ \text{standard}}} (-1)^{\dim(A_M)} i_P^G(r_P^G(\pi))$$

where A_M is the maximal split torus in the center of M .

The Aubert–Zelevinsky involution

Theorem (Aubert 1995)

If $\pi \in \text{Irr}(G)$, there exists $\varepsilon \in \{\pm 1\}$ such that

$$\widehat{\pi} := \varepsilon D_G(\pi) \in \text{Irr}(G).$$

This defines a map, called the **Aubert–Zelevinsky involution**:

$$\begin{array}{ccc} \text{Irr}(G) & \longrightarrow & \text{Irr}(G) \\ \pi & \longmapsto & \widehat{\pi} \end{array}$$

- The map $\pi \mapsto \hat{\pi}$ is an **involution** on $\text{Irr}(G)$.

Key properties

- The map $\pi \mapsto \hat{\pi}$ is an **involution** on $\text{Irr}(G)$.
- If π is **cuspidal**, then $\hat{\pi} = \pi$.

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- If π is **cuspidal**, then $\widehat{\pi} = \pi$.
- It exchanges the **trivial** and the **Steinberg**.

$$\widehat{\mathbf{1}_G} = \text{St}_G \quad \text{and} \quad \widehat{\text{St}_G} = \mathbf{1}_G$$

The case of GL_n

We denote by $\times$ the parabolic induction for GL_n .

If $\pi_i \in \text{Rep}(GL_{n_i})$, then

$$\pi_1 \times \pi_2 = i_{GL_{n_1} \times GL_{n_2}}^{GL_{n_1+n_2}} (\pi_1 \boxtimes \pi_2).$$

We write $\text{Cusp}(GL) = \bigsqcup_{n \geq 1} \text{Cusp}(GL_n)$ for the set of all cuspidal representations.

Definition

A **segment** is a subset of $\text{Cusp}(\text{GL})$ of the form

$$\Delta = \{\rho| \cdot |^x, \rho| \cdot |^{x+1}, \dots, \rho| \cdot |^y\}$$

where $\rho \in \text{Cusp}(\text{GL})$, and $x \leq y$ with $y - x \in \mathbb{N}$.

We use the notation $\Delta = [x, y]_\rho$.

Example: $\Delta = [0, 2]_\rho = \{\rho, \rho| \cdot |, \rho| \cdot |^2\}$

Operations on segments

For a segment $\Delta = [x, y]_\rho$ (with ρ unitary):

- $e(\Delta) = y$ (the **end** of Δ)
- $b(\Delta) = x$ (the **beginning** of Δ)
- $c(\Delta) = \frac{x+y}{2}$ (the **center** of Δ)

Let $\Delta^- = [x, y-1]_\rho$ (removing the last element)

Example: For $\Delta = [0, 2]_\rho$:

- $e(\Delta) = 2, b(\Delta) = 0, c(\Delta) = 1$
- $\Delta^- = [0, 1]_\rho = \{\rho, \rho | \cdot | \}$

Definition

A **multisegment** is a finite formal sum of segments:

$$\mathfrak{m} = \Delta_1 + \cdots + \Delta_r.$$

We denote by Mult the set of all multisegments.

Example: $\mathfrak{m} = [2, 3]_\rho + [2, 3]_\rho + [0, 2]_\rho + [2, 2]_\rho$ is a multisegment.

Theorem (Zelevinsky 1980)

To every segment $\Delta = [x, y]_\rho$, we can associate two irreducible representations $Z(\Delta)$ and $L(\Delta)$:

- $Z(\Delta)$ is the unique irreducible **subrepresentation** of the induced representation

$$\rho| \cdot |^x \times \cdots \times \rho| \cdot |^y$$

- $L(\Delta)$ is the unique irreducible **quotient** of this induced representation.

Theorem (Zelevinsky 1980)

To every multisegment $\mathbf{m} = \Delta_1 + \cdots + \Delta_r$, we associate:

- $Z(\mathbf{m})$: unique irreducible subrepresentation of $Z(\Delta_1) \times \cdots \times Z(\Delta_r)$
- $L(\mathbf{m})$: unique irreducible quotient of $L(\Delta_1) \times \cdots \times L(\Delta_r)$

(satisfying certain technical conditions).

The maps $\mathbf{m} \mapsto Z(\mathbf{m})$ and $\mathbf{m} \mapsto L(\mathbf{m})$ give **bijections**

$$\text{Mult} \xrightarrow{\sim} \text{Irr}(\text{GL}).$$

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Fact: $\widehat{Z(\mathbf{m})} = L(\mathbf{m})$

Question

Given a multisegment $\mathbf{m}$, find $\mathbf{m}^t$ such that

$$\widehat{Z(\mathbf{m})} = Z(\mathbf{m}^t).$$

An explicit algorithm was given by [Mœglin and Waldspurger](#).

Reduction to cuspidal lines

The algorithm works on each **cuspidal line** independently.

Definition

A **cuspidal line** is

$$\mathbb{Z}_\rho = \{\rho | \cdot |^n, n \in \mathbb{Z}\}$$

for a fixed cuspidal representation $\rho \in \text{Cusp}(\text{GL})$.

Let Mult_ρ = multisegments supported on $\mathbb{Z}_\rho$.

Then $\text{Mult} = \bigoplus_\rho \text{Mult}_\rho$ (where the sum runs over cuspidal lines), and this decomposition is compatible with duality:

$$\mathfrak{m} = \sum_\rho \mathfrak{m}_\rho \implies \mathfrak{m}^t = \sum_\rho \mathfrak{m}_\rho^t$$

Definition (Order on segments)

For segments on the same cuspidal line:

$$[x, y]_{\rho} \leq [x', y']_{\rho} \quad \text{if} \quad \begin{cases} x < x' \\ \text{or } x = x' \text{ and } y \geq y' \end{cases}$$



This is NOT the lexicographic order!

Example: $[0, 2]_{\rho} \leq [1, 6]_{\rho} \leq [1, 3]_{\rho}$

The Møglin–Waldspurger algorithm

Let $\mathfrak{m} \in \text{Mult}_\rho$ be a multisegment.

Step 1: Find $e_{\max} = \max\{e(\Delta) \mid \Delta \in \mathfrak{m}\}$.

Step 2: Let Δ_1 be the largest segment (for $\leq$) with $e(\Delta_1) = e_{\max}$.

Step 3: Define inductively $\Delta_1 \geq \Delta_2 \geq \dots \geq \Delta_l$ where Δ_{j+1} is the largest segment such that:

- $\Delta_{j+1} \leq \Delta_j$
- $e(\Delta_{j+1}) = e(\Delta_j) - 1$

(we stop if no such segment exist)

The Mœglin–Waldspurger algorithm

Having found $\Delta_1, \dots, \Delta_l$, define:

- The **first segment of the dual**:

$$\Delta'_1 = [e(\Delta_l), e(\Delta_1)]_\rho$$

- A **reduced multisegment**:

$$\mathfrak{m}^\# = \mathfrak{m} - \sum_{j=1}^l (\Delta_j - \Delta_j^-)$$

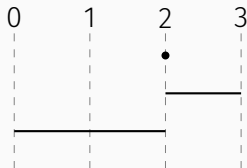
Theorem (Mœglin–Waldspurger 1986)

The dual multisegment is given recursively by:

$$\mathfrak{m}^t = \Delta'_1 + (\mathfrak{m}^\#)^t$$

Example 1

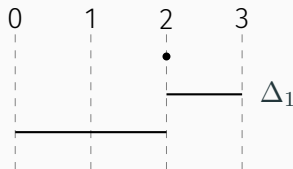
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Dual multisegment: $m^t = \dots$

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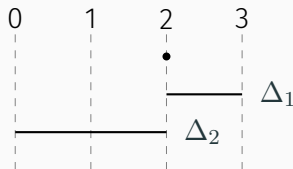
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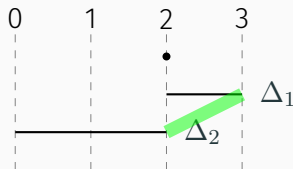
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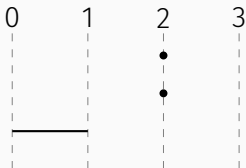
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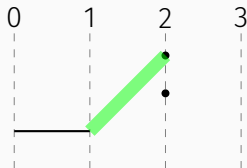
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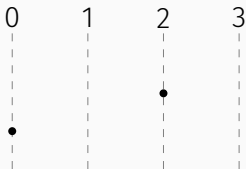
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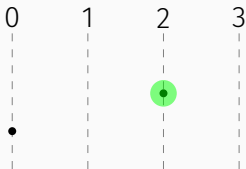
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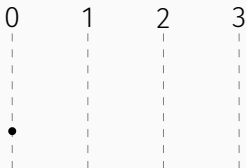
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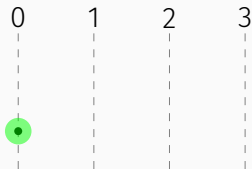
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Compute the dual of $m = [2, 3] + [0, 2] + [2, 2]$.



Dual multisegment: $m^t = [2, 3] + [1, 2] + [2, 2] + [0, 0]$

Example 1

[illegible]

Dual multisegment: $\mathbf{m}^t = [2, 3] + [1, 2] + [2, 2] + [0, 0]$

Example 2: Transpose of a partition

Let $n_1 \geq \dots \geq n_k$ be a partition. Consider the multisegment $\mathbf{m}$ with segments of lengths $n_1, \dots, n_k$ and starting points decreasing by 1:



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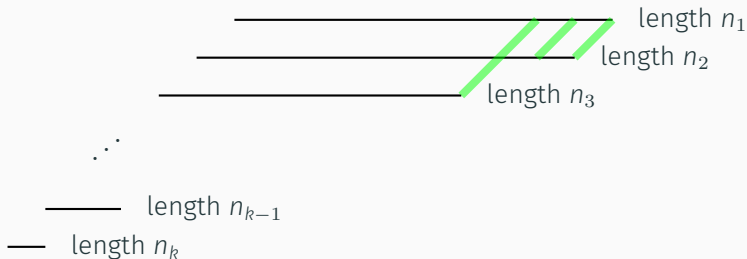
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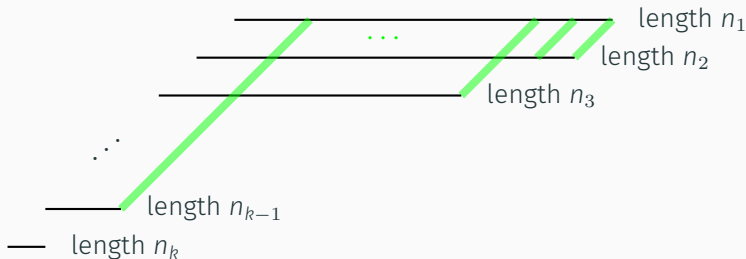
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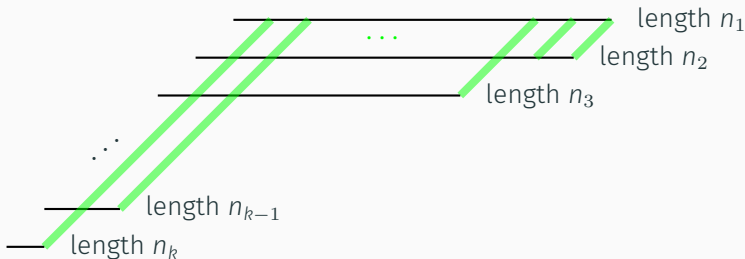
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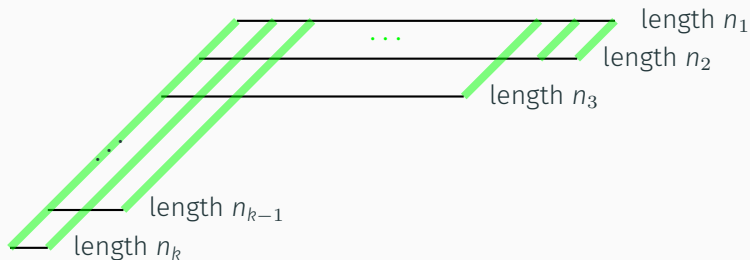
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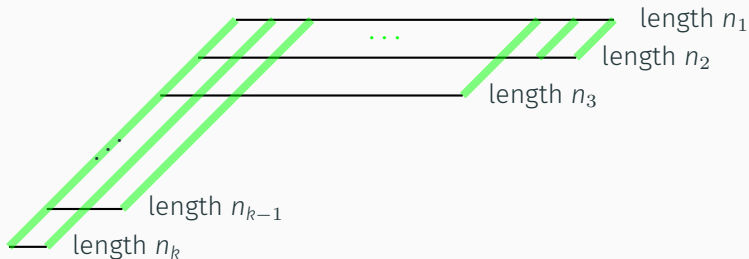
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Then $\mathbf{m}^t$ corresponds to the **transpose partition** (n_i^*) .

The case of classical groups

Classification of irreducible representations

We now consider $G = \mathrm{Sp}_{2n}(F)$ or $\mathrm{SO}_{2n+1}(F)$.

Classification via local Langlands correspondence

Irreducible representations of G are parametrized by **Langlands data**

$$\pi = L(\mathfrak{m}, \phi, \eta)$$

Components of the Langlands data:

- **Multisegment:** $\mathfrak{m} = [x_1, y_1]_{\tau_1} + \cdots + [x_r, y_r]_{\tau_r}$, where $x_i + y_i < 0$
- **Langlands parameter:** $\phi = \bigoplus_{i=1}^k \rho_i \boxtimes S_{a_i}$, where $\rho_i \in \mathrm{Cusp}(\mathrm{GL})$ and $a_i \in \mathbb{N}^*$
- **Character:** η assigns signs ± 1 to each $\rho_i \boxtimes S_{a_i}$

(satisfying technical conditions)

Symmetrization of the data

- For $\Delta = [x, y]_\rho$, define $\Delta^\vee = [-y, -x]_{\rho^\vee}$
- By linearity, it extends to Mult

Example: For $\mathbf{m} = [-1, 2] + [-1, 1]$ we have $\mathbf{m}^\vee = [-2, 1] + [-1, 1]$

Let $\text{Symm} = \{\mathfrak{s} \in \text{Mult} \mid \mathfrak{s}^\vee = \mathfrak{s}\}$ (symmetric multisegments).

Define Symm^ε (symmetric multisegments with signs) as pairs $(\mathfrak{s}, \varepsilon)$ where:

- $\mathfrak{s} \in \text{Symm}$
- ε assigns signs ± 1 to centered segments (those with $c(\Delta) = 0$)

From Langlands data to symmetric multisegment:

$$\begin{aligned}\mathrm{Data}(G) &\longrightarrow \mathrm{Symm}^\varepsilon \\ (\mathfrak{m}, \phi, \eta) &\longmapsto (\mathfrak{s}, \varepsilon)\end{aligned}$$

where the symmetric multisegment is

$$\mathfrak{s} = \sum_{\Delta \in \mathfrak{m}} (\Delta + \Delta^\vee) + \sum_{\rho \boxtimes S_a \in \phi} \left[-\frac{a-1}{2}, \frac{a-1}{2} \right]_\rho$$

and the signs are given by $\varepsilon \left(\left[-\frac{a-1}{2}, \frac{a-1}{2} \right]_\rho \right) = \eta(\rho \boxtimes S_a)$

We write $\pi = L(\mathfrak{s}, \varepsilon)$ for this symmetrized form.

The AD algorithm

We define an involution

$$\text{AD} : \text{Symm}^\varepsilon \longrightarrow \text{Symm}^\varepsilon$$

Theorem (L.-Mínguez)

Let $\pi = L(\mathfrak{s}, \varepsilon) \in \text{Irr}(G)$. Then

$$\widehat{\pi} = L(\text{AD}(\mathfrak{s}, \varepsilon)).$$

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Remark: It is not easy to find an involution on Symm^ε !

Three types of cuspidal representations

- AD is defined on each cuspidal line $\mathbb{Z}_\rho$
- Unlike GL_n , the cuspidal ρ matters!
- Three cases depending on ρ : ugly, bad, good

AD is defined **recursively**:

For $(\mathfrak{s}, \varepsilon) \in \text{Symm}^\varepsilon$, construct:

- $(\mathfrak{s}_1, \varepsilon_1)$: first piece of the dual
- $(\mathfrak{s}^\#, \varepsilon^\#)$: reduced symmetric multisegment

such that

$$\text{AD}(\mathfrak{s}, \varepsilon) = (\mathfrak{s}_1, \varepsilon_1) + \text{AD}(\mathfrak{s}^\#, \varepsilon^\#)$$

AD algorithm (good case)

Let $(\mathfrak{s}, \varepsilon) \in \text{Symm}^\varepsilon$.

Step 1: Order segments by a specific order $\preceq$.

Step 2: Construct descending chain $\Delta_1 \succeq \cdots \succeq \Delta_r$ where:

- Δ_1 is the largest segment with maximal end
- Δ_{j+1} is the largest segment such that:
 1. $\Delta_{j+1} \preceq \Delta_j$
 2. $e(\Delta_{j+1}) = e(\Delta_j) - 1$
 3. **Sign condition:** if $c(\Delta_{j+1}) = c(\Delta_j) = 0$ then $\varepsilon(\Delta_{j+1})\varepsilon(\Delta_j) = -1$

Step 3: Extract $\mathfrak{s}_1$ from ends of Δ_j and beginnings of $\Delta_j^\vee$.

Step 4: Compute $\mathfrak{s}^\#$ by removing these parts + adjust signs.

Example

Setting: $G = \mathrm{Sp}_{2n}(F)$ and $\rho = 1$ (good case)

Langlands datum:

$$\pi = L([-2, 0] + [-1, -1] + [-1, -1], S_1 + S_1 + S_3, (-1, -1, 1))$$

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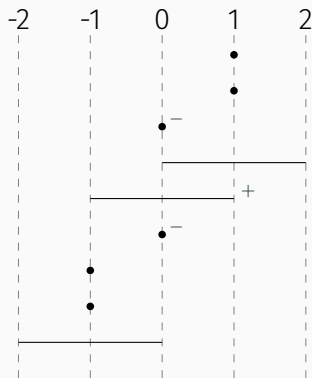
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Symmetrization:

$$\mathfrak{s} = [-2, 0] + [0, 2] + [-1, -1] + [1, 1] + [-1, -1] + [1, 1] + [0, 0] + [0, 0] + [-1, 1]$$

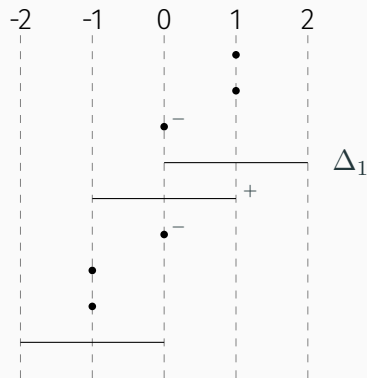
with $\varepsilon([0, 0]) = -1$ and $\varepsilon([-1, 1]) = 1$.

Example: Step by step



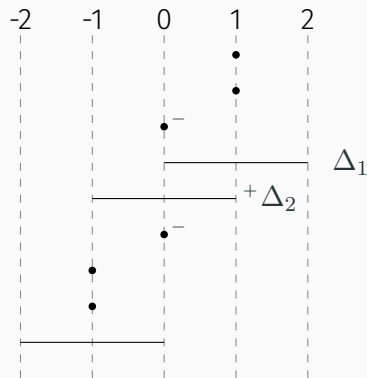
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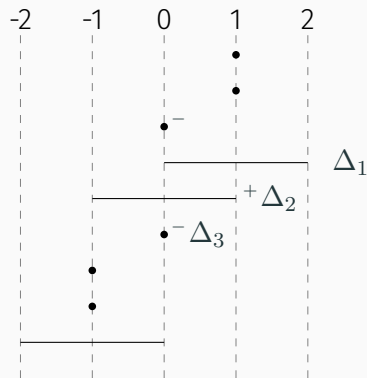
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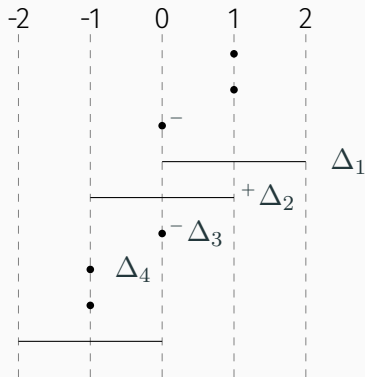
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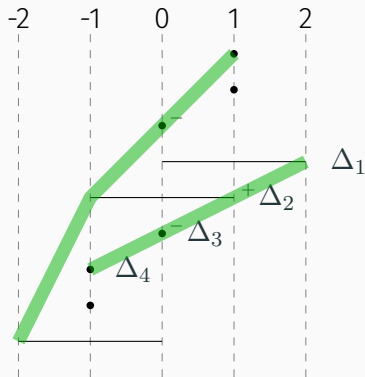
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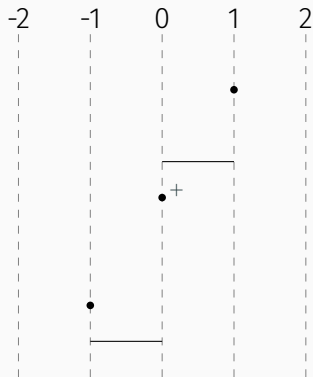
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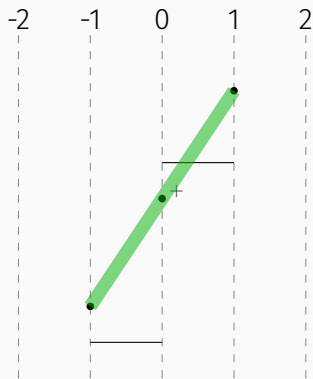
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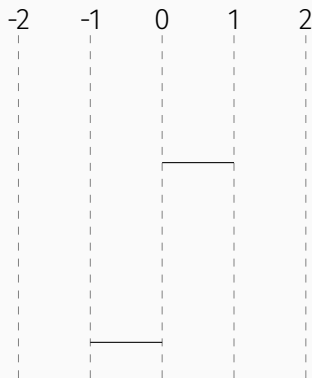
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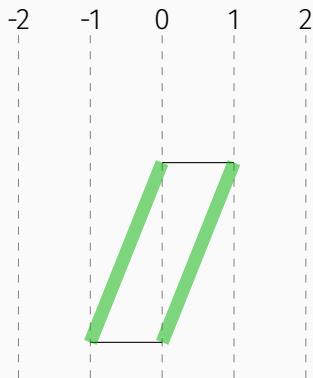
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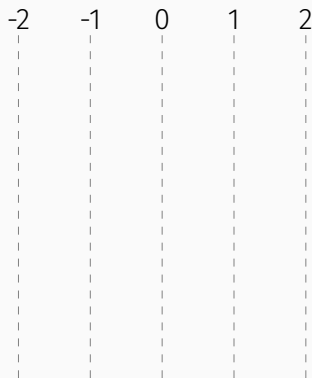
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Example: Result

Input:

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Converting back to Langlands data:

$$\hat{\pi} = L([-2, 1] + [-1, 0], S_3, (1))$$

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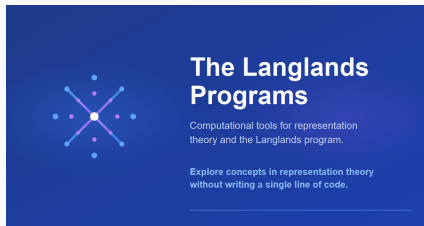
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Implementation

Implementations:

- **Python/Sage package:**
`github.com/ThomasLanard/aubert-zelevinsky-duality`
- **Web application** (in development with Petar Bakić and Elad Zelingher):
`langlandsprograms.com`



Machine learning

Motivation:

- AI excels at finding patterns in large datasets.
- In mathematics, generating large and precise datasets is straightforward.
- AI can be used to uncover new relations between mathematical objects.

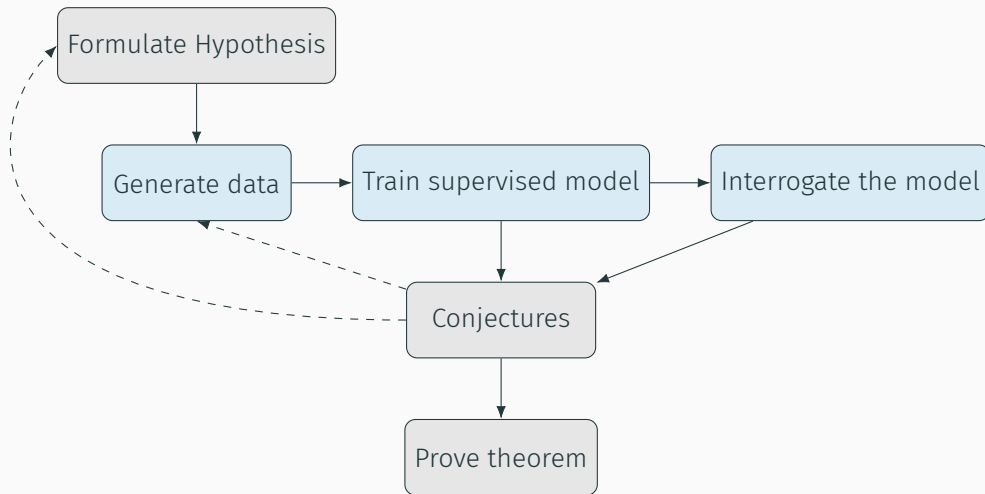
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In our work:

- We use **supervised learning**, relying on the Atobe–Mínguez algorithm, to **gain intuition** about the Aubert–Zelevinsky duality.
- We began this process **without making any mathematical assumptions or conjectures**, in order to explore what structures the machine learning model might reveal on its own.

Machine learning workflow



A **neural network** is a function $\varphi : \mathbb{R}^n \rightarrow \mathbb{R}^m$ given by composition:

$$\varphi = \varphi_r \circ \cdots \circ \varphi_1$$

where each layer is:

$$\varphi_i : \mathbb{R}^{n_i} \xrightarrow{\text{linear}} \mathbb{R}^{n_{i+1}} \xrightarrow{\text{ReLU}} \mathbb{R}^{n_{i+1}}$$

ReLU (Rectified Linear Unit): $x \mapsto \max(0, x)$ applied componentwise

(no ReLU on the last layer φ_r)

Supervised learning

Goal: Approximate an unknown function $\hat{\varphi} : \mathbb{R}^n \rightarrow \mathbb{R}^m$

Training data: Samples $(x_i, \hat{\varphi}(x_i))_{i=1, \dots, N}$

Loss function: Measures prediction error

$$l(\varphi) = \frac{1}{N} \sum_{i=1}^N \|\varphi(x_i) - \hat{\varphi}(x_i)\|^2$$

Training: Use **gradient descent** to find parameters that minimize $l(\varphi)$

First attempt: Predict the entire dual

Question

Can we predict the entire dual $AD(\mathbf{m}, \phi, \eta)$ directly?

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Result

The model shows **poor accuracy**.

New strategy: Instead of predicting the entire dual at once, try to predict **one segment**.

Question

Which segment should we predict first?

Thank you for your attention!